

a library, absorbing the rhythm and flow of words. You then begin to write an essay and try to figure out the best word to pick after the word 'blue'. Based on the text you have read, your mind races through options you have seen before—ocean, skies, night, steel, eyes—each a possible successor. But your choice is informed by the symphony of words that came before as well as the context in which these words were written. Similarly, these algorithms predict the next word by considering a multitude of predecessors, then selecting from a shortlist of the most likely next words. Hence, if the preceding paragraphs are about the environment, then the likely candidate for the next word is perhaps 'ocean' or 'skies'. But which one do you pick? These algorithms often pick the next word from such shortlists at random! And this randomness manifests as a remarkably creative composition (question for another day—so is creativity some kind of randomness?). This is the secret sauce behind the seemingly spontaneous and intelligent responses of tools like ChatGPT.

These advancements have empowered machines to not just consume but understand and generate text in a way that simulates human thought. The key to this prowess lies in AI's ability to learn complex relationships between words from an ocean of text—web pages, books, speeches, even source code. When transformers are trained on such large swathes of text, they are termed as large language models (LLMs). But it doesn't stop there. The realm of LLMs' learning extends to images and videos because the words in the captions are used to learn correlations between the words and pixels. This versatility means that these LLMs can now engage with almost any form of human knowledge, be it legal documents, historical records or artistic expressions, if we can convert them into strings of words, with their innate patterns.

Hence, words, the building blocks of our communication, history and knowledge, are now within the grasp of AI. This isn't just a technical leap; it's a fundamental shift in how we interact with information and express ourselves. We have begun to live in a world where AI already passes high school biology or history exams as well as the best students. It helps an author draft an essay, for India's leading magazine, on AI. At the Murty Classical